

Comparing AMM Mechanisms: Slippage, Impermanent Loss, and Deterministic LP Returns

BitDA Final Project

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Abstract

Automated market makers (AMMs) exchange simplicity, capital efficiency, and portfolio exposure in materially different ways. This study implements a reproducible analytical simulator for four designs: Uniswap V2 constant product, Uniswap V3 fixed-range concentrated liquidity, Curve StableSwap, and Balancer weighted pools. We compare nine configurations over three liquidity levels, seven normalized trade sizes, six terminal price changes, and three fixed daily-volume assumptions. StableSwap with amplification $A = 1000$ produces only 0.061% slippage for a trade equal to 10% of total value locked (TVL), while Uniswap V2 produces 16.875%. However, high amplification increases loss during a small depeg. Narrow Uniswap V3 liquidity also minimizes slippage, but its fixed range produces the greatest loss after large price moves. Balancer’s 80/20 pool has worse slippage in the tested trade direction but lower impermanent loss than a 50/50 pool. The results show that AMM selection is a market-design decision rather than a universal ranking.

1 Introduction

An AMM replaces an order book with an invariant that determines how pool inventory and price change after a trade. The invariant makes execution available without a matching counterparty, but liquidity providers (LPs) implicitly sell assets that appreciate and buy assets that depreciate. Different AMMs control this inventory response differently. Constant-product pools are general and permissionless; concentrated liquidity allocates capital to a selected price interval; StableSwap flattens the price curve for correlated assets; and weighted pools encode asymmetric portfolio exposure.

This report answers three questions:

1. How do the mathematical mechanisms change execution slippage?
2. How do terminal prices change LP value relative to holding?
3. Under fixed 30-day volume assumptions, when do fees compensate for impermanent loss?

The contribution is a small, tested simulator rather than a qualitative comparison alone. Every reported result is generated from a versioned YAML scenario file. A Monte Carlo extension measures return distributions and V3 range risk. The study excludes live on-chain data, loss-versus-rebalancing, and active V3 strategies because they require additional market and strategy assumptions.

2 Mathematical Models

2.1 Constant Product

For reserves x and y , Uniswap V2 preserves

$$xy = k. \quad (1)$$

For gross exact input Δx and fee rate f , the effective input is $\Delta x_e = (1 - f)\Delta x$. Output is

$$\Delta y = y - \frac{k}{x + \Delta x_e}. \quad (2)$$

The marginal pre-trade price in units of token y per token x is $p = y/x$. As input grows relative to reserves, execution traverses a more curved portion of the invariant and slippage rises.

2.2 Concentrated Liquidity

Uniswap V3 represents liquidity L between lower and upper square-root prices $\sqrt{p_a}$ and $\sqrt{p_b}$. At a price inside the range, token amounts are

$$x(p) = L \left(\frac{1}{\sqrt{p}} - \frac{1}{\sqrt{p_b}} \right), \quad (3)$$

$$y(p) = L (\sqrt{p} - \sqrt{p_a}). \quad (4)$$

Below the range the position is entirely token x ; above it the position is entirely token y . A narrower interval places more effective liquidity near the current price, but stops earning fees sooner and converts fully into the underperforming side after a sufficiently large move [2].

We compare fixed ranges $[0.95, 1.05]$, $[0.80, 1.20]$, and $[0.50, 2.00]$. These are labeled narrow, medium, and wide. Fixed ranges isolate capital efficiency and range risk; active rebalancing would add strategy, gas, and path-dependent assumptions beyond this comparison.

2.3 StableSwap

Curve’s StableSwap combines constant-sum behavior near parity with constant-product protection farther away. For n assets, balances x_i , amplification A , and invariant D , the invariant is defined implicitly by

$$An^n \sum_i x_i + D = ADn^n + \frac{D^{n+1}}{n^n \prod_i x_i}. \quad (5)$$

The simulator solves D and the post-trade balance iteratively following the whitepaper algorithm [3]. We use equal initial balances and compare $A = 10, 100, 1000$. This logarithmic spread makes the transition from a curved pool toward a flatter near-parity region visible. StableSwap is evaluated for impermanent loss only at $\pm 5\%$ because it is designed for correlated assets, not unrestricted volatile pairs.

2.4 Weighted Constant Mean

Balancer generalizes constant product to

$$x^{w_x} y^{w_y} = k, \quad w_x + w_y = 1. \quad (6)$$

For token x input, output is

$$\Delta y = y \left[1 - \left(\frac{x}{x + \Delta x_e} \right)^{w_x/w_y} \right]. \quad (7)$$

The 50/50 pool is mathematically equivalent to constant product. The 80/20 configuration represents an asymmetric portfolio and is a common use case for retaining greater exposure to one asset [4]. Comparing both separates the effect of the invariant family from the effect of weights.

3 Methodology

3.1 Normalization and Metrics

All pools begin at price $p_0 = 1$ and equal total capital. For weighted pools, reserves follow their target weights, so an 80/20 pool starts with $x = 0.8$ TVL and $y = 0.2$ TVL. This makes its

marginal price one while preserving intended exposure. V3 liquidity L is scaled so that the fixed-range position has the same initial value as every other pool.

Slippage compares the gross-input execution price with the pre-trade marginal price:

$$s = 1 - \frac{\Delta y / \Delta x}{p_0}. \quad (8)$$

Using gross input means the reported value includes both price impact and the trading fee. Fee paid is also stored separately in the generated CSV.

Impermanent loss (IL) compares the arbitrated terminal pool value with holding the pool’s own initial inventory:

$$\text{IL}(p_T) = \frac{V_{\text{pool}}(p_T)}{x_0 p_T + y_0} - 1. \quad (9)$$

This is essential for a fair 80/20 comparison: its baseline is holding 80/20, not holding 50/50. StableSwap terminal balances are found by minimizing pool value along the invariant at the external price, equivalent to reaching the arbitrage tangent.

3.2 Scenario Selection

The slippage experiment uses TVLs of \$100 thousand, \$1 million, and \$10 million. These orders of magnitude demonstrate absolute liquidity scaling. Trade input ranges from 0.1% to 20% of TVL. Normalized size exposes invariant shape, while the absolute-input liquidity plot demonstrates that a fixed trade experiences less impact in a larger pool.

Terminal price changes are -50% , -20% , -5% , $+5\%$, $+20\%$, and $+50\%$. They cover small movement, ordinary volatility, and a large stress move without introducing a stochastic process. LP returns use daily volume equal to 10%, 50%, or 100% of TVL over 30 days. The middle case is used in the main return figure.

3.3 Deterministic LP Return

Fee return as a fraction of initial capital is approximated as

$$r_f = v_d f \times 30, \quad (10)$$

where v_d is daily volume divided by TVL. Let $F = ar_f V_0$ be the available fee value. We report both absolute LP return

$$r_{\text{abs}} = \frac{V_{\text{pool}}(p_T) + F}{V_0} - 1 \quad (11)$$

Table 1: Approved configurations and rationale.

Model	Config.	Purpose
V2	0.30% fee	General constant-product baseline
V3	narrow, medium, wide	Isolate efficiency versus range risk
StableSwap	$A = 10, 100, 1000$	Weak-strong amplification
Weighted	50/50, 80/20	Baseline versus asymmetric exposure

and excess return versus holding:

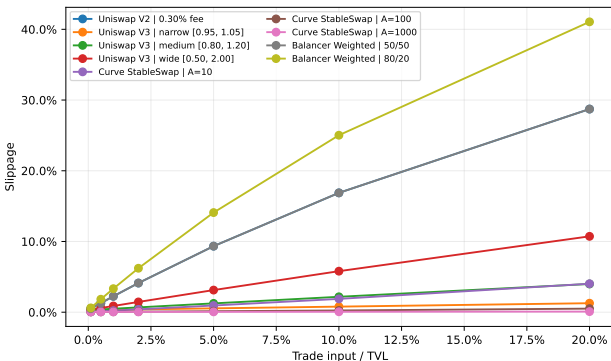
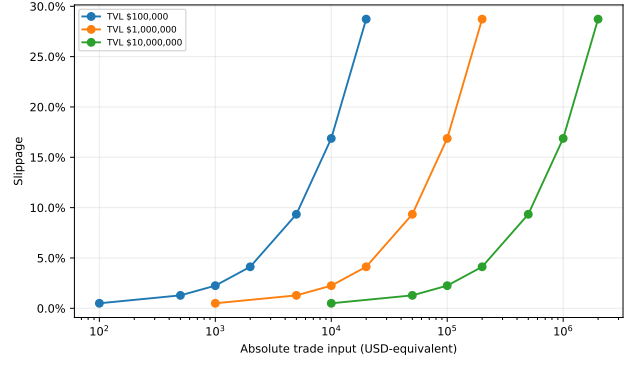
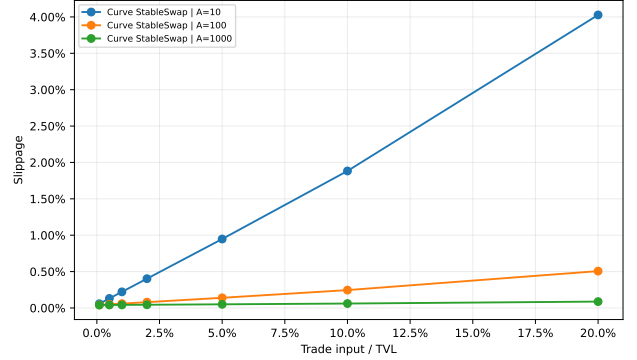
$$r_{\text{excess}} = \frac{V_{\text{pool}}(p_T) + F}{x_0 p_T + y_0} - 1. \quad (12)$$

The availability factor a is one for full-range models. For V3 it is the fraction of a linear path from p_0 to p_T that remains inside the fixed range. This is deliberately transparent, but it is not a claim that real volume is uniform over time or price. Comparative figures use excess return because it isolates whether providing liquidity outperformed holding the same initial assets. A positive excess return does not imply a positive absolute investment return.

4 Slippage Results

Figure 1 shows that invariant choice strongly changes execution. At a 10%-of-TVL trade, V2 and Balancer 50/50 both produce 16.875% slippage, confirming their mathematical equivalence. Balancer 80/20 produces 25.026% in the tested direction because token x is already the heavily weighted reserve; withdrawing token y traverses a steeper effective curve. Reversing direction would change this comparison, so the result is not a general statement that 80/20 pools always have worse execution.

V3 demonstrates the benefit of concentrating

**Figure 1:** Slippage versus normalized trade size at \$1 million TVL.**Figure 2:** A fixed absolute trade receives lower V2 slippage as TVL increases.**Figure 3:** Amplification reduces StableSwap slippage near parity.

capital. At the same 10%-of-TVL input, narrow, medium, and wide ranges produce 0.789%, 2.179%, and 5.802% slippage. Their advantage is conditional on the trade remaining near the active interval. A sufficiently large input reaches the lower bound, after which this single-position model supplies no additional liquidity. The simulator rejects such an exact-input quote rather than reporting a partial fill as a complete trade.

StableSwap is the most efficient around parity. At 10% input, slippage falls from 1.883% at $A = 10$ to 0.246% at $A = 100$ and 0.061% at $A = 1000$. Increasing A flattens the near-parity curve, but the later impermanent-loss result shows why this efficiency is not free.

Figure 2 separates normalized shape from absolute scale. For a given trade/TVL ratio, constant-product slippage is identical across liquidity levels. For a fixed dollar-equivalent trade, however, a larger pool moves less because the trade represents a smaller reserve fraction. This explains why “high liquidity” must be defined relative to intended order size.

5 Impermanent-Loss Results

Figure 4 exposes the opposite side of capital efficiency. At a +50% terminal move, V2 and Bal-

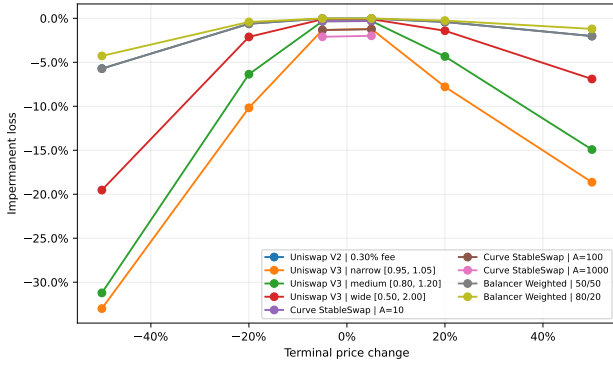


Figure 4: Impermanent loss relative to each configuration’s initial holdings.

ancer 50/50 lose 2.020% relative to holding. Balancer 80/20 loses 1.203%, because its invariant trades less of the initially dominant token for this direction and its benchmark preserves the same 80/20 exposure. At -50% , the losses are larger: 5.719% for 50/50 and 4.275% for 80/20. The asymmetry occurs because equal percentage increases and decreases do not represent reciprocal prices.

Fixed-range V3 positions have much stronger range risk. At $+50\%$, narrow, medium, and wide positions lose 18.635%, 14.919%, and 6.898%. At -50% the losses become 32.998%, 31.217%, and 19.526%. A narrow position exchanges its inventory aggressively inside a small interval and then becomes entirely one token. Its superior local liquidity therefore creates a larger opportunity cost when price leaves the range.

StableSwap is shown only around parity. For a $+5\%$ move, IL rises in magnitude from 0.311% at $A = 10$ to 1.245% at $A = 100$ and 1.988% at $A = 1000$. Strong amplification maintains a near-one internal price longer, allowing arbitrageurs to remove more of the appreciating asset during a depeg. High A is appropriate only when the economic assumption of correlation is credible.

6 LP-Return Results

At 50% daily volume, a 0.30% fee produces fee value equal to 4.5% of initial capital before V3 availability adjustment. V2 excess return versus HODL is 3.677% at $+20\%$, 1.580% at $+50\%$, and 0.281% at -50% . In the last case, the LP still has a negative 24.789% absolute return because both initially held assets lost value, but fees make liquidity provision slightly better than HODL.

Balancer 80/20 has the highest excess return among volatile-pair full-range models in these tested directions because it receives the same assumed fee rate while experiencing lower IL. Its

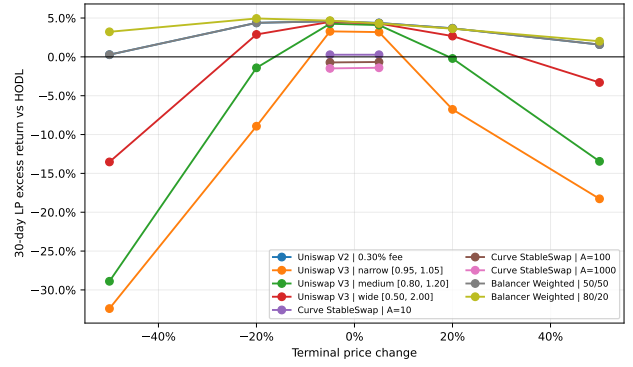


Figure 5: Deterministic 30-day LP excess return versus HODL at daily volume equal to 50% of TVL.

excess return is 3.623% at $+20\%$ and 2.012% at $+50\%$. This conclusion depends directly on equal volume and fee assumptions; actual order flow may differ by pool design.

V3 outcomes depend on both IL and time in range. At $+20\%$, the wide range earns 2.677% excess return, while the medium and narrow ranges lose 0.209% and 6.764%. The narrow range earns fees for only one quarter of the assumed linear path before price crosses its upper bound. The result demonstrates that headline capital efficiency is not the same as realized excess return versus HODL.

StableSwap charges only 0.04% in this model, giving a 0.6% gross 30-day fee return at 50% daily volume. This covers the $A = 10$ small-depeg loss but not the $A = 100$ or $A = 1000$ loss. A highly amplified pool therefore requires either stronger confidence in the peg, more volume, a higher fee, or additional risk controls.

7 Volume Sensitivity

The fixed-volume scenarios make the break-even dependence explicit. Table 2 reports excess return versus HODL at a $+20\%$ terminal move. V2 and both weighted pools remain positive even at 10% daily volume because their IL is small in this scenario. The V3 wide position requires more than 10% daily volume to break even, while the medium position requires close to the 50% case. The narrow position remains negative even at daily volume equal to TVL because it is active for only one quarter of the assumed price path and its IL dominates collected fees.

Volume sensitivity also clarifies why the return analysis is not a forecast. The simulator assigns the same volume fraction to mechanisms with very different execution quality. In a real market, low slippage may attract more flow, while fragmented liquidity, incentives, routing, and fee tiers deter-

Table 2: 30-day excess return versus HODL at +20% price change.

Configuration	10%	50%	100%
V2 / Weighted 50/50	0.40%	3.68%	7.77%
Weighted 80/20	0.52%	3.62%	7.50%
V3 narrow	-7.58%	-6.76%	-5.74%
V3 medium	-3.51%	-0.21%	3.92%
V3 wide	-0.60%	2.68%	6.77%

mine which LPs receive that flow. The controlled assumption is useful because it isolates fee-rate and availability effects, but deployment analysis must replace it with observed or modeled order flow.

8 Monte Carlo Risk Analysis

The deterministic scenarios isolate specific terminal prices, but LPs face a distribution of price paths. As additional analysis, we generate 10,000 zero-drift geometric Brownian motion (GBM) paths for each annualized volatility level $\sigma \in \{20\%, 60\%, 100\%\}$. Each path begins at one and contains 30 daily steps:

$$P_{t+1} = P_t \exp\left(-\frac{1}{2}\sigma_d^2 + \sigma_d Z_t\right), \quad (13)$$

where $\sigma_d = \sigma/\sqrt{365}$ and Z_t is standard normal.

Zero drift avoids embedding a directional market forecast. The three volatility levels represent low, moderate, and high volatile-asset markets. Thirty daily observations match the base return period, while 10,000 paths provide stable distribution summaries at modest computational cost. The fixed seed 20260611 makes the output exactly reproducible.

StableSwap is excluded from this extension because unrestricted GBM does not represent stable-asset mean reversion or jump-depeg risk. A credible StableSwap stochastic analysis requires a separate price process.

For each path, terminal IL uses the same analytical valuation as the deterministic analysis. Full-range models earn the assumed 0.30% fee on daily volume equal to 50% of TVL. A V3 position earns fees only for the fraction of observed daily prices inside its fixed range. A fixed position can earn fees again if price re-enters; no active rebalancing occurs.

Figure 6 separates average outcome from downside risk. At $\sigma = 100\%$, V2 and weighted 50/50 have a mean excess return of 3.56% and a fifth-percentile excess return of 0.58%. Weighted 80/20 improves these to 4.04% and 1.69% relative to its

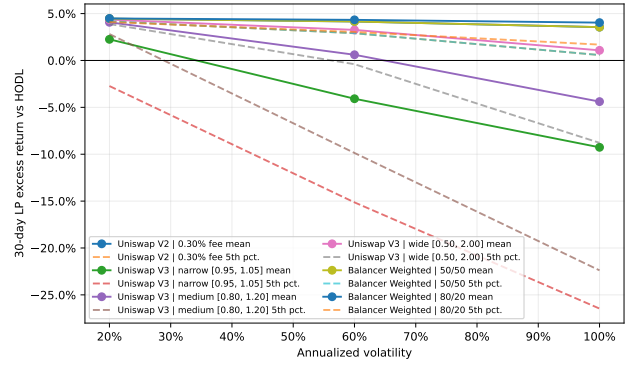


Figure 6: Monte Carlo mean and fifth-percentile 30-day LP excess returns versus HODL.

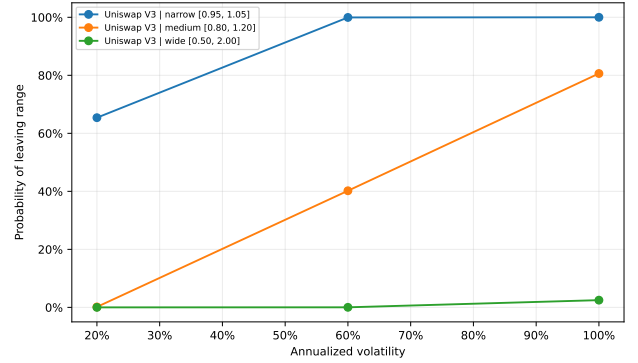


Figure 7: Probability that a fixed V3 position leaves its range at least once.

weighted hold benchmark. This is consistent with its lower IL sensitivity in the tested exposure.

V3 outcomes depend strongly on range width. Narrow V3 has a positive 2.25% mean excess return at $\sigma = 20\%$, but its fifth percentile is -2.73% and its underperformance probability is 18.37%. At $\sigma = 100\%$, its mean excess return falls to -9.26% and underperformance probability reaches 85.23%. Medium V3 also turns negative on average at high volatility. Wide V3 retains a positive 1.08% mean excess return at $\sigma = 100\%$, but its fifth percentile is -8.78%, showing that a positive mean does not eliminate material downside.

Figure 7 explains the fee and return differences. Narrow V3 leaves its range on 65.39% of low-volatility paths and essentially every moderate- or high-volatility path. Its mean in-range fraction falls from 79.70% to 23.60%. Medium V3 exits on 40.20% of paths at $\sigma = 60\%$ and 80.62% at $\sigma = 100\%$. Wide V3 exits on only 2.49% of high-volatility paths.

The stochastic extension reinforces rather than replaces the deterministic conclusion: concentrated liquidity improves local execution, but realized LP performance depends on the probability and duration of range exit. GBM is still a simplification; it excludes jumps, volatility clustering,

endogenous volume, intraday crossings, and correlations between price movement and order flow.

9 Mechanism Comparison and Market Fit

No configuration dominates every metric. StableSwap and narrow V3 both increase local capital efficiency by making stronger assumptions about where price should remain. StableSwap assumes correlation; V3 LPs choose a bounded interval. When those assumptions fail, inventory changes more aggressively than in a general full-range pool.

V2 is comparatively inefficient for small trades because liquidity is spread from zero to infinity, but it is robust in the limited sense that liquidity never becomes inactive. Its simple invariant also provides a transparent baseline for auditing other designs. Balancer extends this baseline by allowing the pool itself to represent a portfolio preference. That can reduce IL relative to the corresponding weighted hold position, but it changes directional depth and should not be judged with a 50/50 benchmark.

For traders, the relevant question is whether active liquidity exists for the intended trade and direction. For LPs, the relevant question is whether fee income compensates for inventory rebalancing under plausible price paths. These perspectives can conflict: the configuration offering traders the least slippage can expose LPs to the strongest adverse selection.

10 Limitations

The base scenarios are deliberately deterministic, and the Monte Carlo extension uses a deliberately simple price process. Results should be interpreted as controlled mechanism comparisons, not forecasts.

- Fee income is proportional to an assumed daily volume and does not model endogenous volume, fee compounding, or competition among LPs.
- Terminal IL is path independent for the full-range analytical models, while actual fee earnings and V3 inventory management are path dependent.
- Monte Carlo paths use zero-drift GBM with constant volatility. They exclude jumps, volatility clustering, and price-volume dependence.
- V3 uses one fixed position and a linear-path availability factor. It excludes ticks shared by other LPs, active repositioning, gas, and rebalancing losses. Exact-input quotes beyond the fixed position’s capacity are rejected rather than routed through additional positions.
- StableSwap uses two equally valued assets and an ideal arbitrage equilibrium. It excludes admin fees, imbalance fees, withdrawal behavior, and irreversible depeg dynamics.
- Balancer tests token x input only. Weighted pools have intentionally asymmetric depth, so reverse-direction execution must be evaluated separately for a deployment decision.
- Smart-contract precision, integer rounding, oracle behavior, gas, and protocol-specific implementation details are outside the analytical model.

These limits are also why the report avoids a single “best AMM” score. Combining metrics into one number would require subjective probabilities for volume, volatility, depeg, range management, and trade direction.

11 Reproducibility and Validation

The implementation separates invariant modules from shared metrics and experiment orchestration. Automated tests cover known outputs, invariant preservation, V3 inventory below/inside/above its range, StableSwap amplification behavior, zero-change IL, deterministic scenario counts, and figure generation. Monte Carlo tests cover fixed-seed determinism, zero-volatility behavior, complete summaries, and ordered V3 range risk. The approved YAML configuration expands to 189 slippage rows, 42 IL rows, 126 LP-return rows, and 18 Monte Carlo summary rows. Return outputs include both absolute LP return and excess return versus HODL; comparative figures and underperformance probabilities use the latter.

The experiment command regenerates all CSV files and PDF figures. Monte Carlo uses NumPy’s random number generator with a fixed seed, and no external API is used. This makes every table and figure repeatable from the repository state and allows individual assumptions to be changed without rewriting model code.

Reproduction requires three commands from the repository root: `run pytest`; `run scripts/run_experiments.py` with `PYTHONPATH=src`; then compile `report/main.tex`

Table 3: Qualitative interpretation of the simulated trade-offs.

Mechanism	Main strength	Main LP risk	Suitable market	Important caveat
V2 constant product	General, continuous full-range liquidity	IL under directional movement	Permissionless volatile pairs	Capital is spread over all prices
V3 fixed range	Very low local slippage	Range exit and one-sided inventory	Pairs with active LP management or bounded expectations	Fixed positions can stop earning fees
StableSwap	Extremely low near-parity slippage	Depeg drains the stronger asset	Credibly correlated or redeemable assets	Higher A magnifies depeg exposure
Weighted pool	Encodes desired portfolio weights	Direction-dependent execution and IL	Treasury, index, and liquidity bootstrap-ping designs	Comparison must preserve weighted hold benchmark

Table 4: Selected validation claims and automated checks.

Claim	Check
V2 and weighted 50/50 are equivalent	Equal output and invariant tests
Swaps remain on their invariant	Pre/post numerical tolerance tests
V3 becomes one-sided outside range	Below, inside, and above inventory tests
Higher A reduces near-parity slippage	Ordered quote comparison
IL uses the correct hold benchmark	Zero-change and known V2 formula tests
Experiments are reproducible	Exact row counts and deterministic frames
Monte Carlo is reproducible	Fixed seed and zero-volatility path tests

with Tectonic. Generated CSV files remain available for auditing values that are summarized or rounded in this report. Source code and reproduction instructions are available at <https://github.com/Yu-20001/bitda-final>.

12 Conclusion

AMM design shifts risk rather than eliminating it. Constant product offers a general full-range baseline but produces material slippage for large reserve fractions. Concentrated liquidity improves execution dramatically near the current price, at the cost of strong range-dependent LP outcomes. StableSwap achieves the lowest near-parity slippage in the study, but high amplification turns a small depeg into a meaningful LP loss. Weighted pools preserve asymmetric portfolio exposure and can reduce IL relative to that weighted benchmark, while producing direction-dependent execution.

Monte Carlo results make the range trade-off distributional: narrow V3 loses fee availability and

develops strongly negative downside as volatility rises, while full-range and wide-range positions retain more stable average outcomes.

For general volatile markets without active LP management, V2-like full-range liquidity remains the simplest robust choice. For bounded or actively managed markets, V3 can use capital much more efficiently. For credibly redeemable or correlated assets, StableSwap offers superior execution when amplification is selected conservatively. Weighted pools are most useful when portfolio exposure is itself part of the protocol objective. The appropriate mechanism therefore follows from market assumptions, LP objectives, and acceptable failure modes.

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